



Primera División

Country of origin	Frequency	Percentage
Europe	280	70
North America	16	4
South America	56	14
Africa	32	8
Asia	16	4
Total	400	100



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Europe	280	70
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Asia	16	4
Total	400	100



Primera División

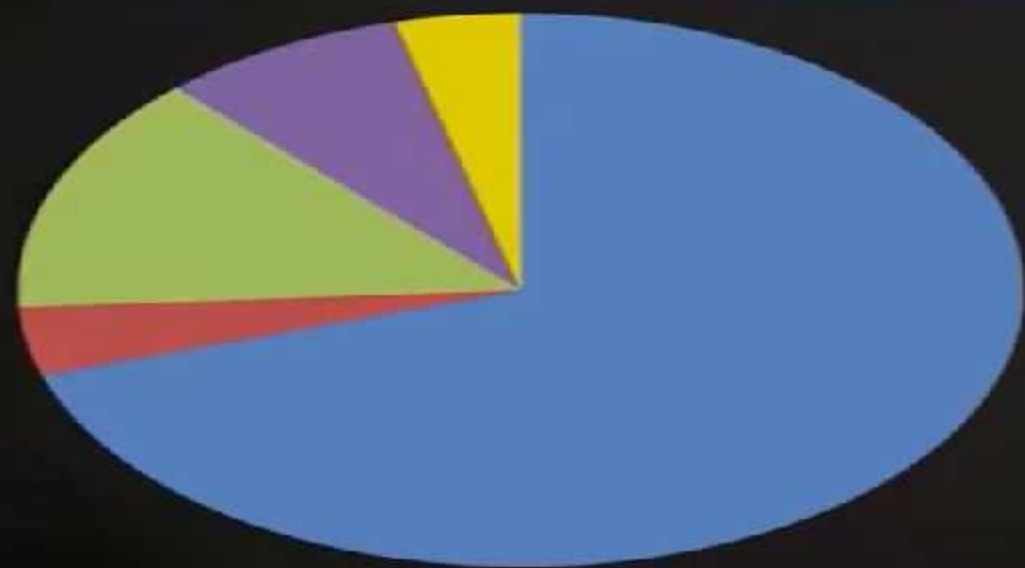
Country of origin	Frequency	Percentage
Europe	280	70
North America	16	4
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Asia	16	4
Total	400	100



Primera División

Country of origin	Frequency	Percentage
Europe	280	70
North America	16	4
South America	56	14
Africa	32	8
Asia	16	4
Total	400	100

PIE CHART



Europe

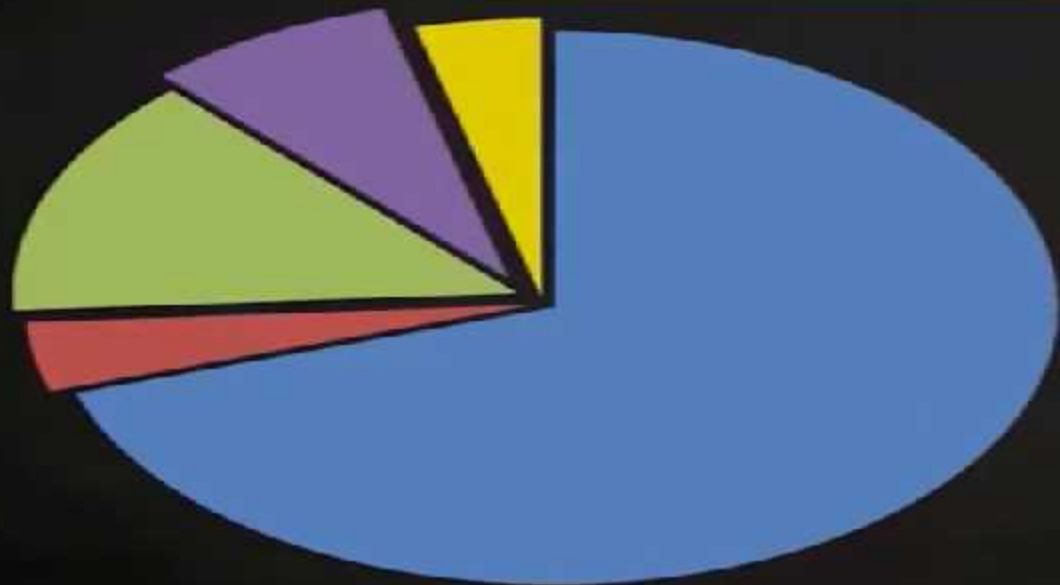
North America

South America

Africa

Asia

PIE CHART



Europe

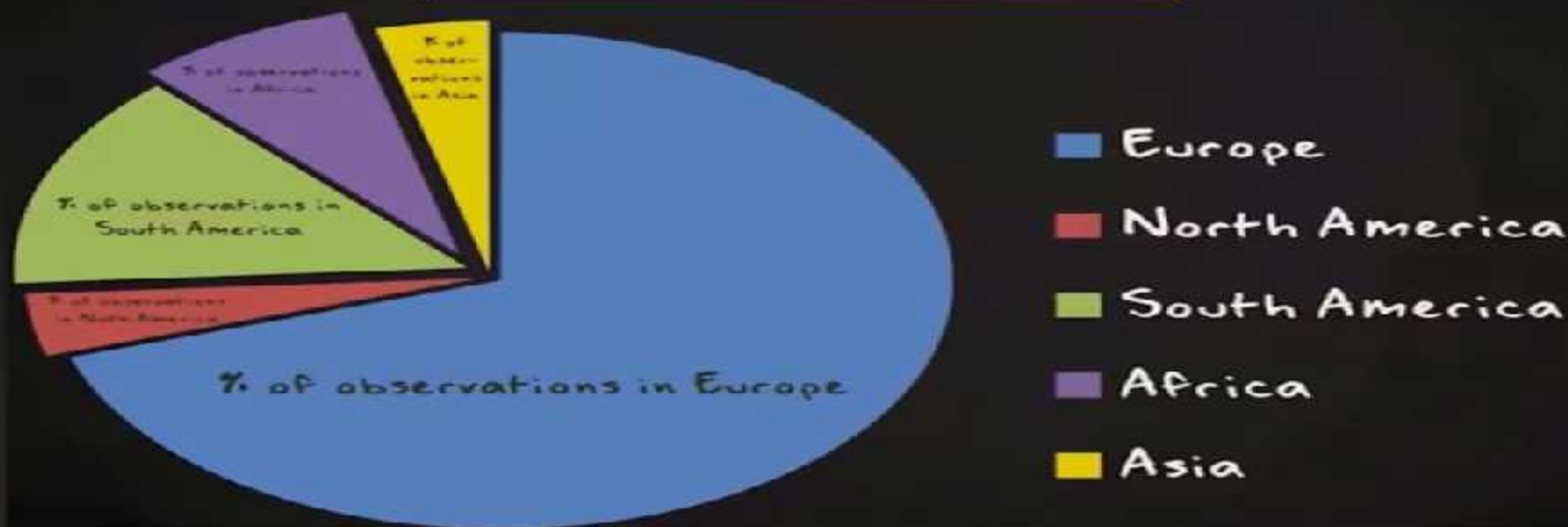
North America

South America

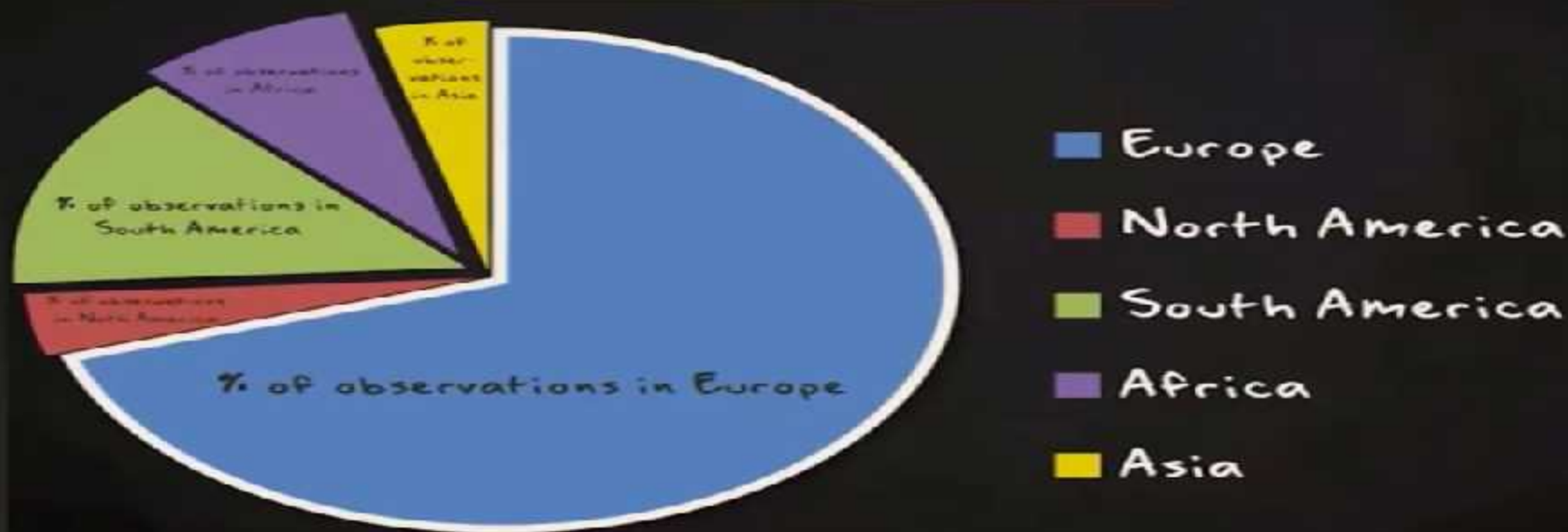
Africa

Asia

PIE CHART



PIE CHART



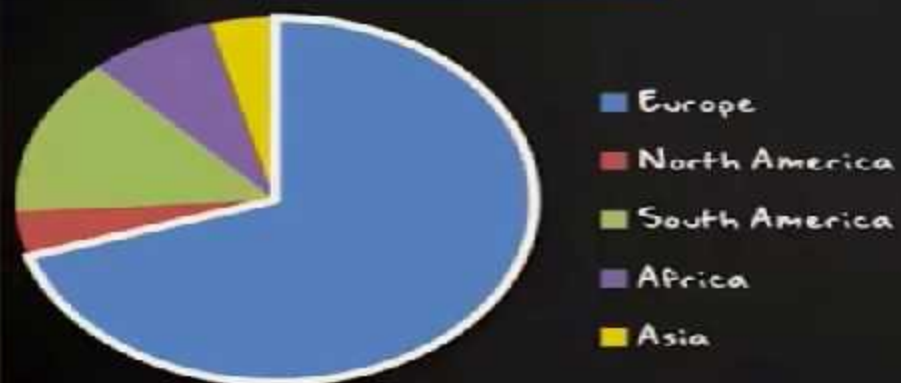
BAR GRAPH



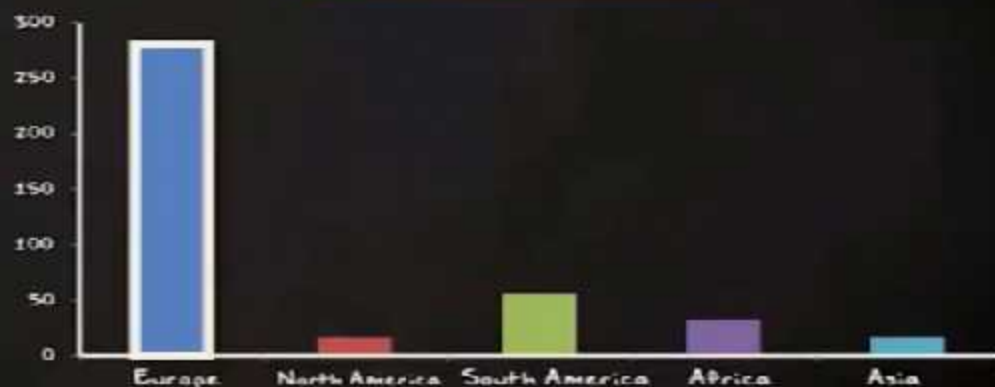
BAR GRAPH



PIE CHART



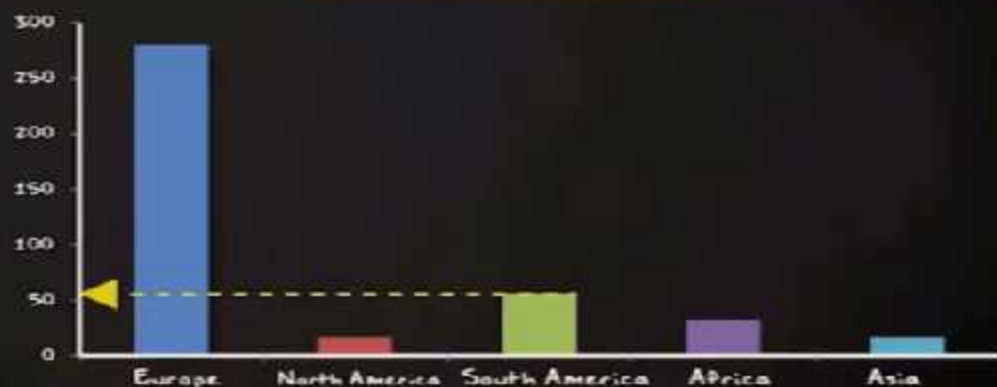
BAR GRAPH



PIE CHART



BAR GRAPH



PIE CHART



BAR GRAPH

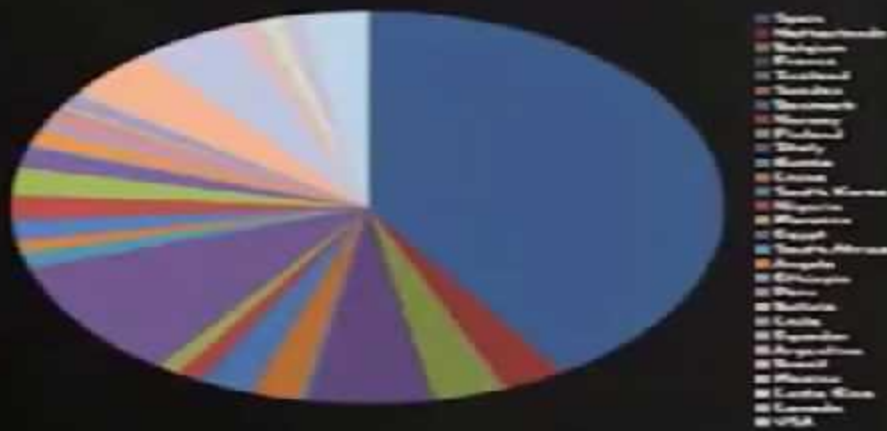
high number of categories

PIE CHART



BAR GRAPH

high number of categories

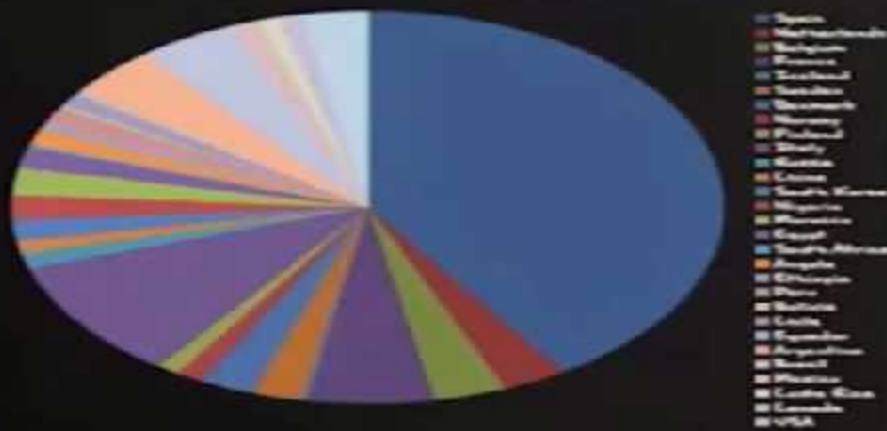


PIE CHART



BAR GRAPH

high number of categories



Dot plot

Dot plot

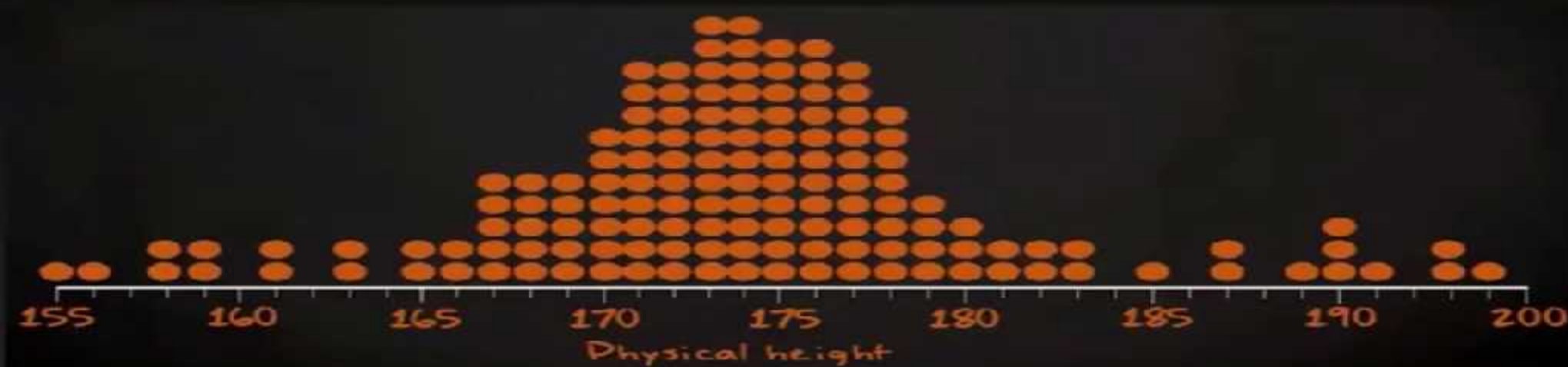
	Physical height
Player 1	176
Player 2	180
Player 3	165
Player 4	177
Player 5	167
Player 6	170
Player 7	175
Player 8	178
Player 9	174
Player 10	172

Dot plot

	Physical height
Player 1	176
Player 2	180
Player 3	165
Player 4	177
Player 5	167
Player 6	170
Player 7	175
Player 8	178
Player 9	174
Player 10	172



Dot plot



HISTOGRAM

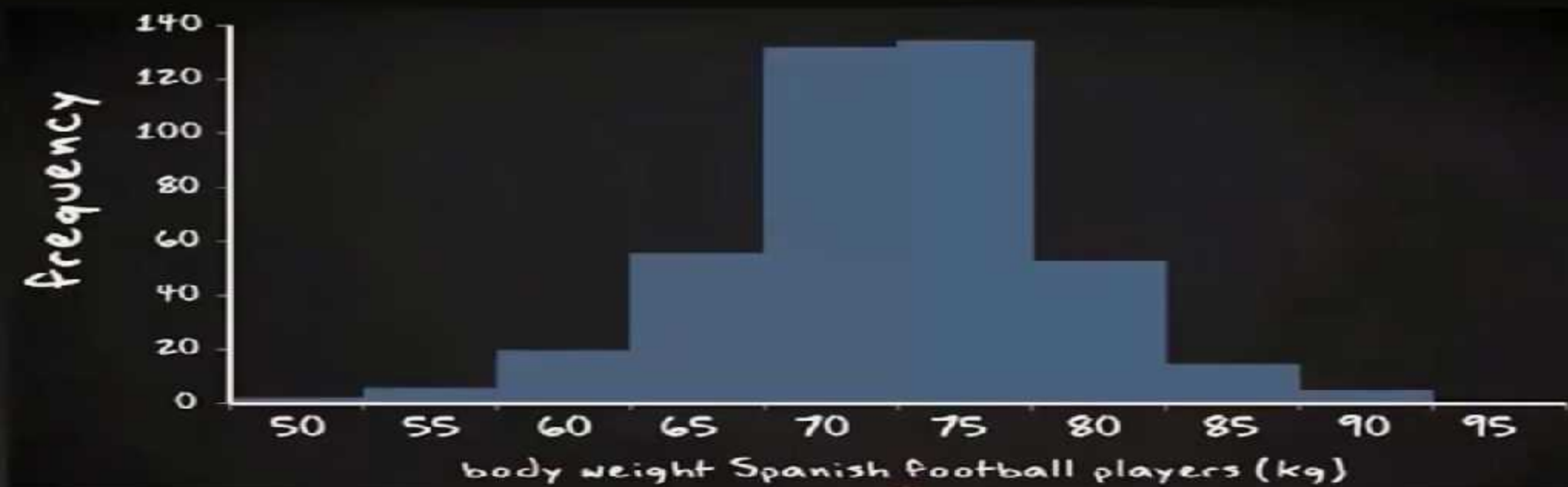


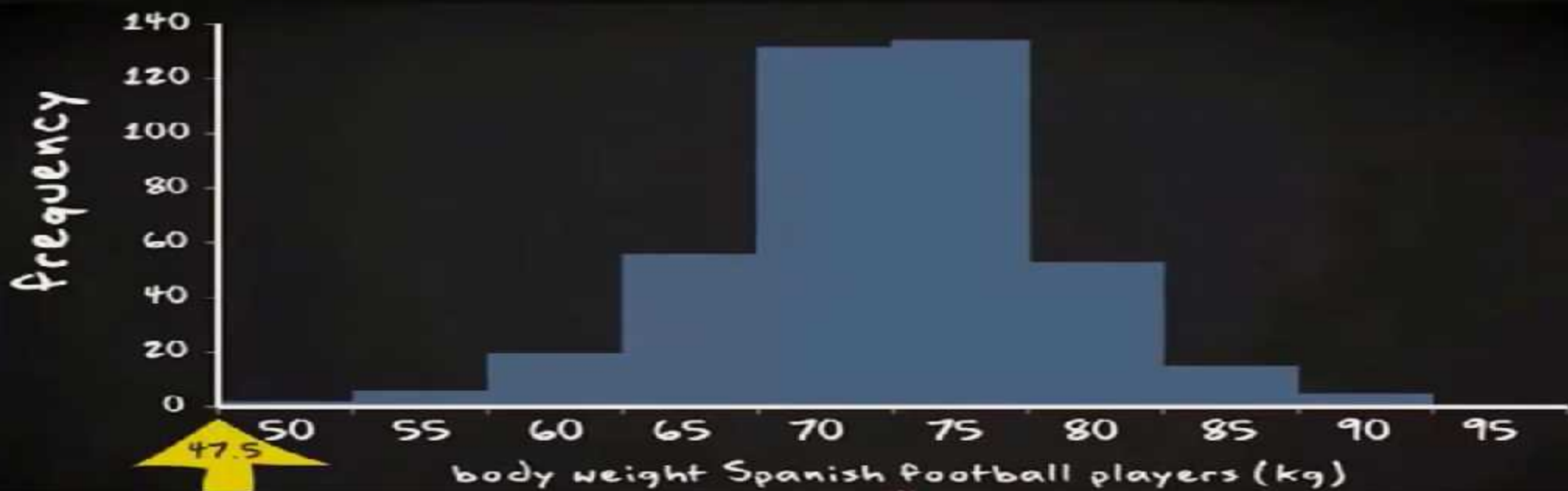
Body weights

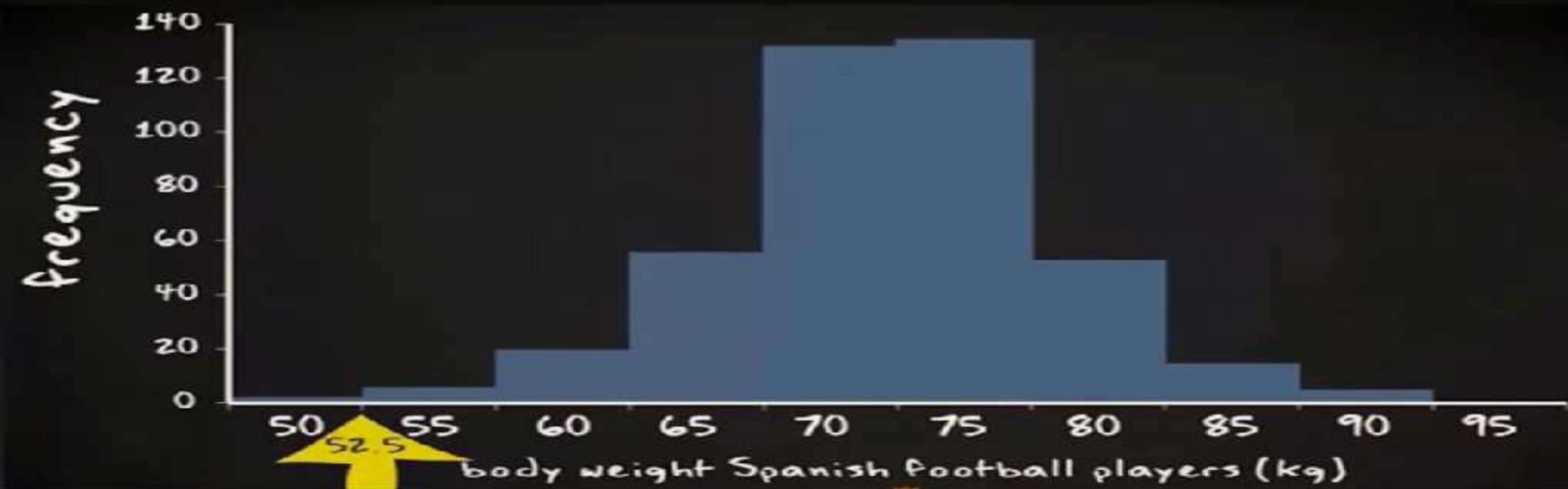
Body weights

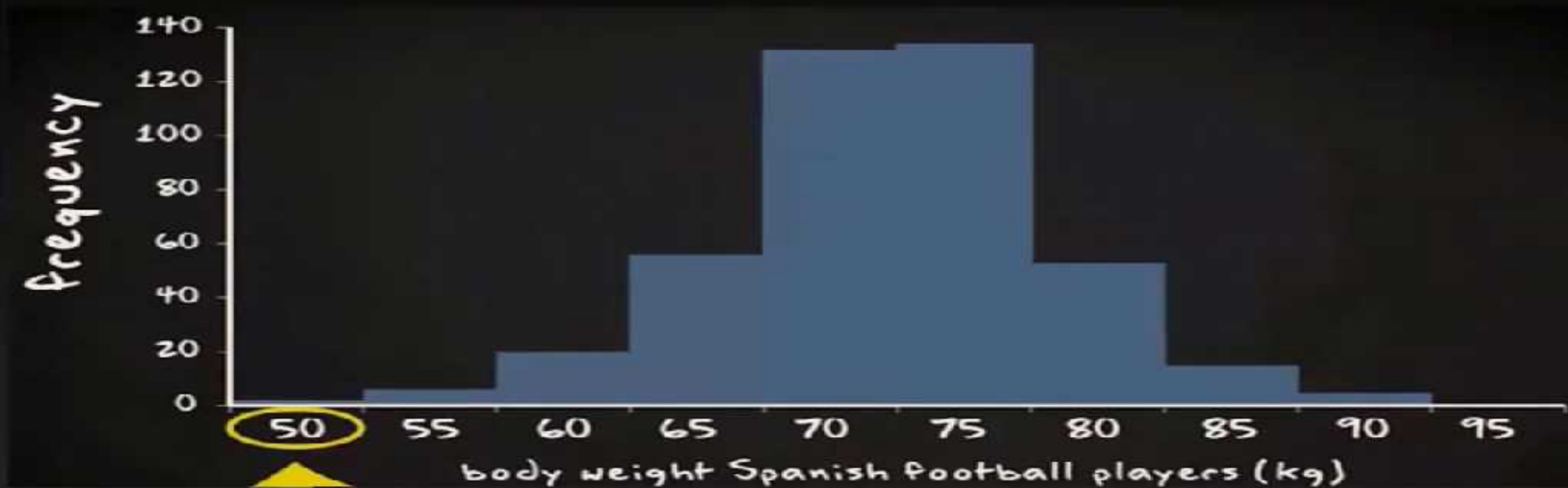
83.9 kg

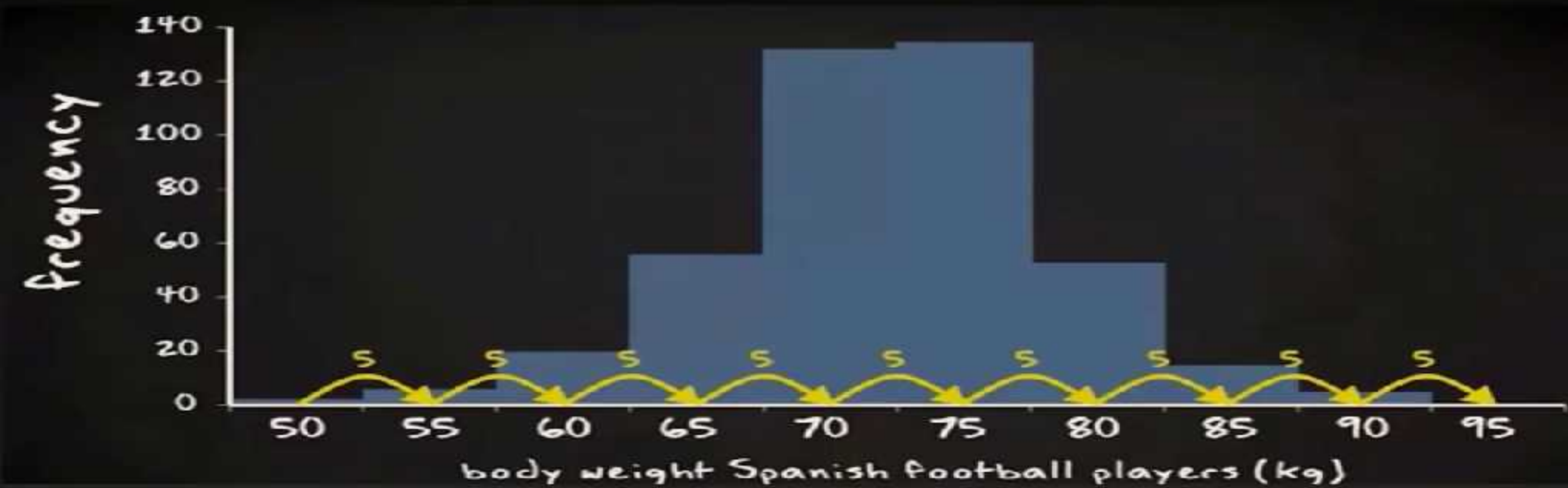
74.5 kg

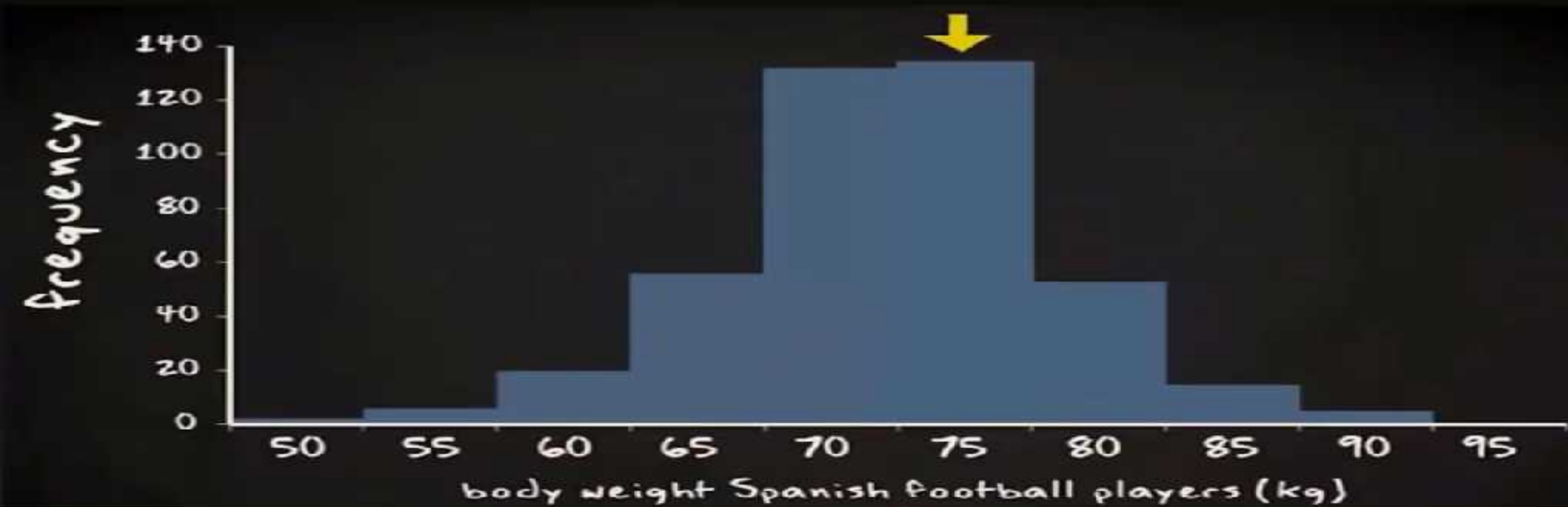


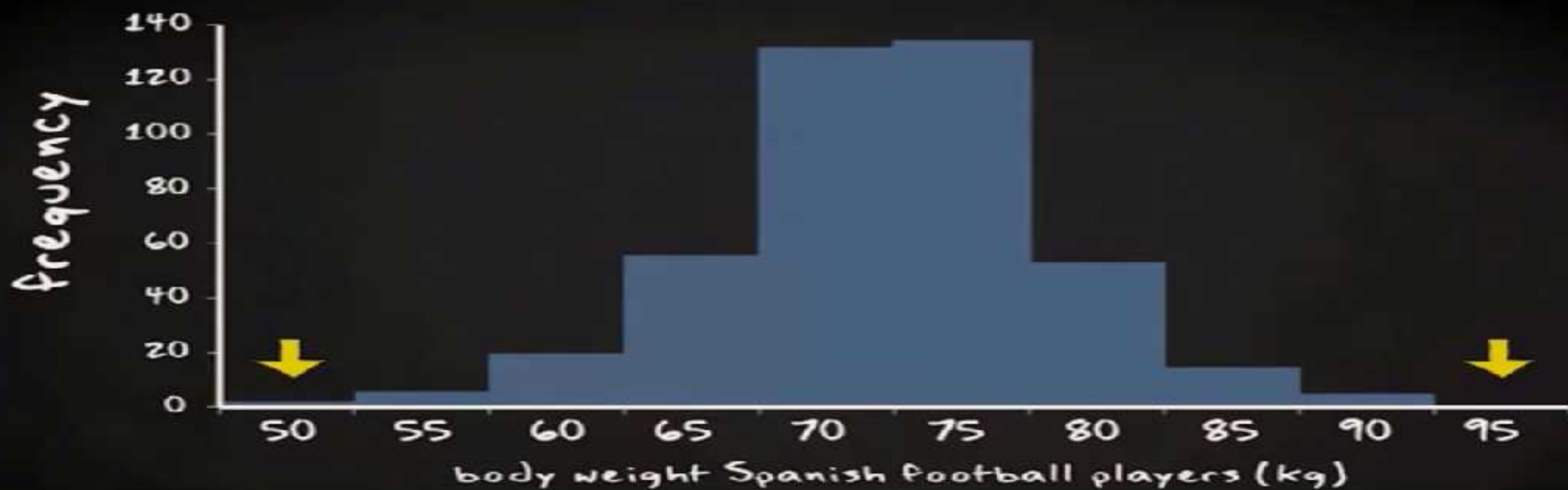


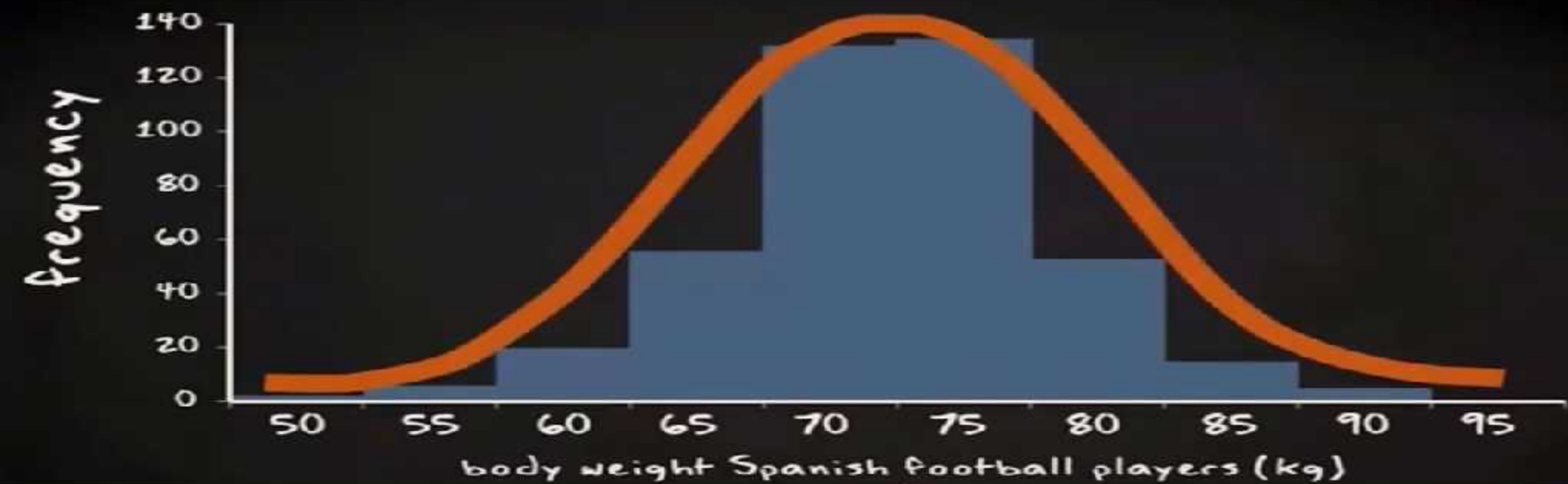
















bell shaped



skewed to the right



bell shaped



skewed to the right



skewed to the left



bell shaped



skewed to the right



skewed to the left

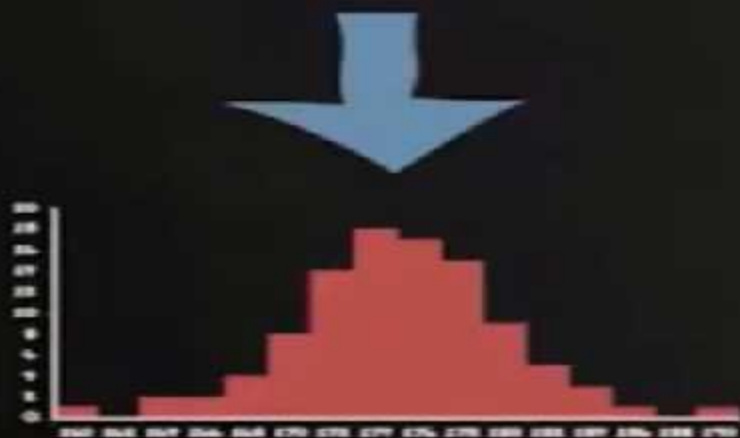


two peaks

nominal/
ordinal



interval/
ratio



summarizing
a distribution

```
graph TD; A[summarizing a distribution] --> B[GRAPH]; A --> C[CENTER];
```

GRAPH

CENTER

summarizing
a distribution

```
graph TD; A[summarizing a distribution] --> B[GRAPH]; A --> C[CENTER]; C --> D[mode]; C --> E[median]; C --> F[mean];
```

GRAPH

CENTER

mode

median

mean

summarizing
a distribution

```
graph TD; A[summarizing a distribution] --> B[GRAPH]; A --> C[CENTER]; C --> D[mode]; C --> E[median]; C --> F[mean];
```

GRAPH

CENTER

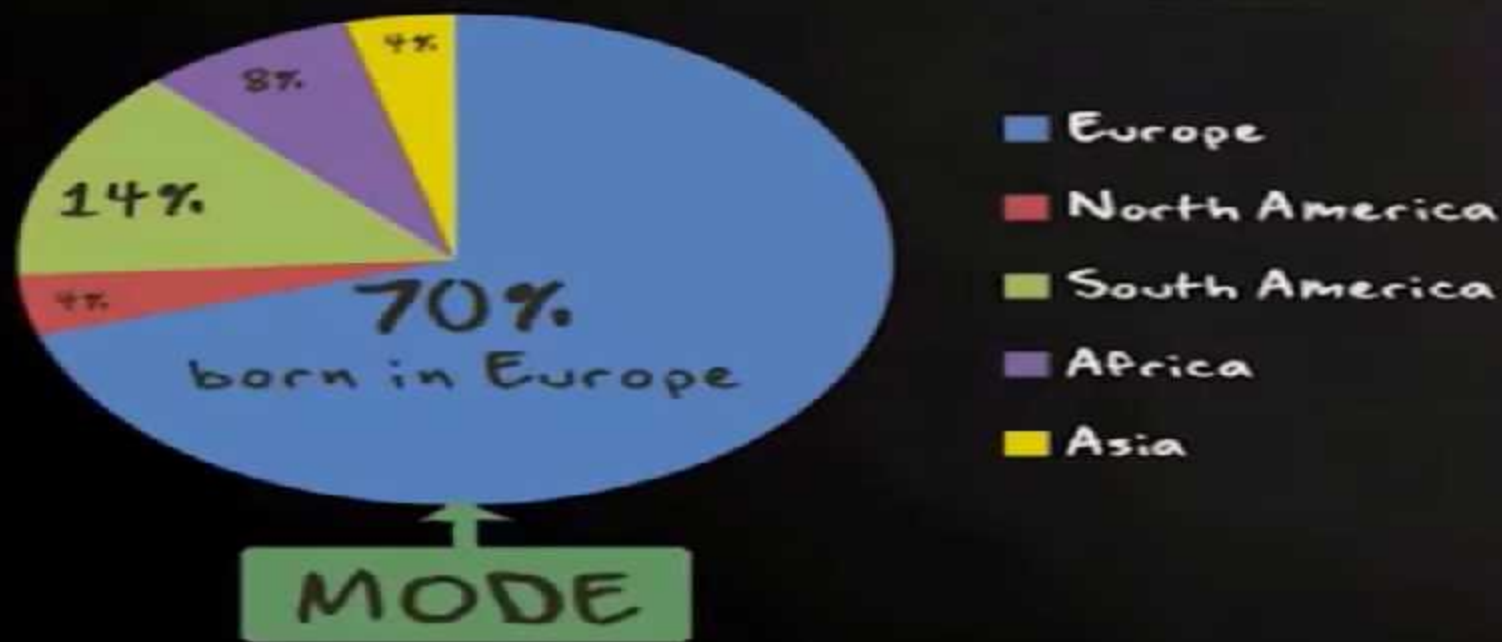
mode

median

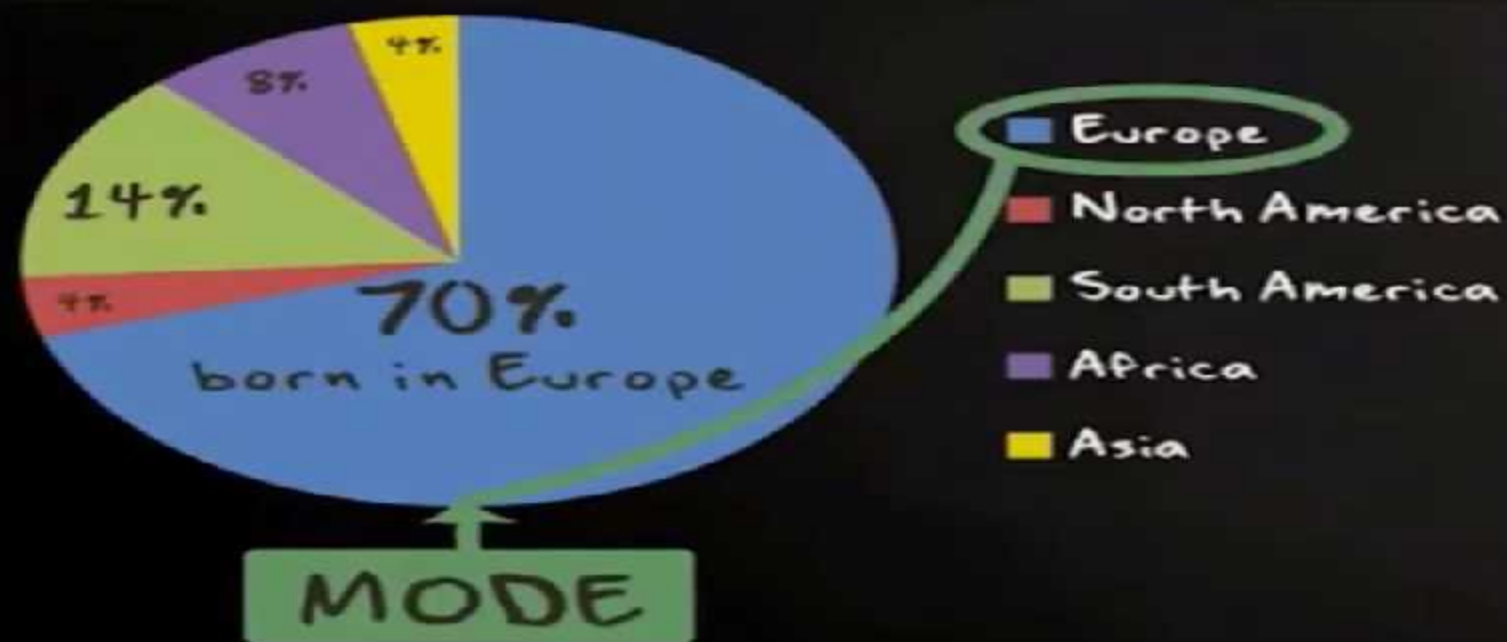
mean

Measures of central tendency

Mode



Mode









Franco Galtón

$N = 500$

How sympathetic do you think Franco Galtón is?



0

1

2

3

4

5

6

7

8

9

10





Franco Galtón

$N = 500$

How sympathetic do you think Franco Galtón is?



0 1 2 3 4 5 6 7 8 9 10





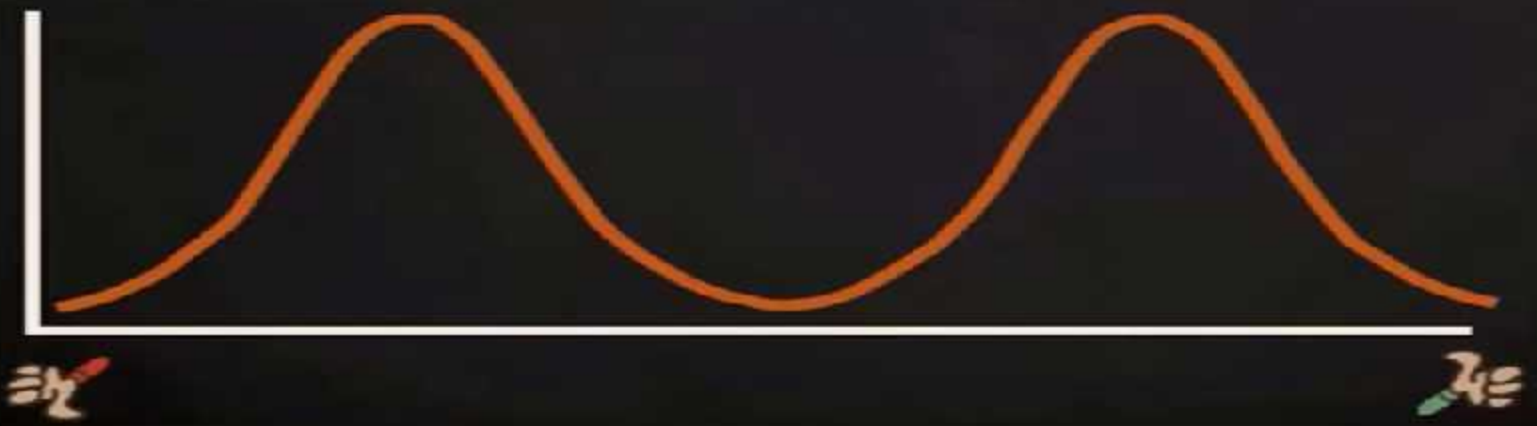
Franco Galtón

$N = 500$

How sympathetic do you think Franco Galtón is?



0 1 2 3 4 5 6 7 8 9 10





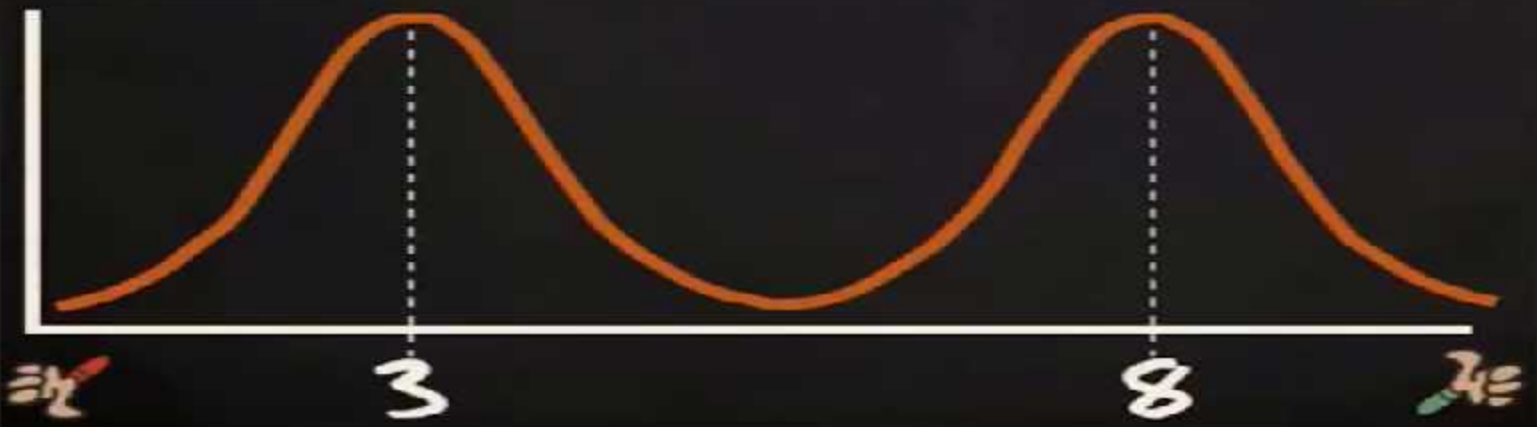
Franco Galtón

$N = 500$

How sympathetic do you think Franco Galtón is?



0 1 2 3 4 5 6 7 8 9 10





Franco Galtón

$N = 500$

How sympathetic do you think Franco Galtón is?



0 1 2 3 4 5 6 7 8 9 10



3

bi-modal distribution

8



Median



Tomás Bayez



How sympathetic do you think Tomás Bayez is?

👉 0 1 2 3 4 5 6 7 8 9 10 👈



How sympathetic do you think Tomás Bayez is?

👉 0 1 2 3 4 5 6 7 8 9 10 👈

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8



How sympathetic do you think Tomás Bayez is?

0 1 2 3 4 5 6 7 8 9 10

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8



How sympathetic do you think Tomás Bayez is?

0 1 2 3 4 5 6 7 8 9 10

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8

MEDIAN

6 7 7 8 8 8 9



How sympathetic do you think Tomás Bayez is?

0 1 2 3 4 5 6 7 8 9 10

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8

MEDIAN
↓
~~6~~ ~~7~~ ~~7~~ 8 ~~8~~ ~~8~~ ~~9~~



How sympathetic do you think Tomás Bayez is?

0 1 2 3 4 5 6 7 8 9 10

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8





How sympathetic do you think Tomás Bayez is?

👉 0 1 2 3 4 5 6 7 8 9 10 👈

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8
Respondent 8	7



How sympathetic do you think Tomás Bayez is?

0 1 2 3 4 5 6 7 8 9 10

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8
Respondent 8	7

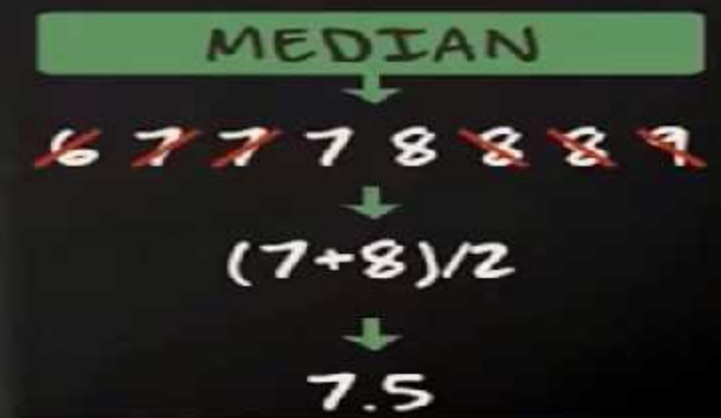
MEDIAN
↓
6 7 7 7 8 8 8 9



How sympathetic do you think Tomás Bayez is?

0 1 2 3 4 5 6 7 8 9 10

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8
Respondent 8	7

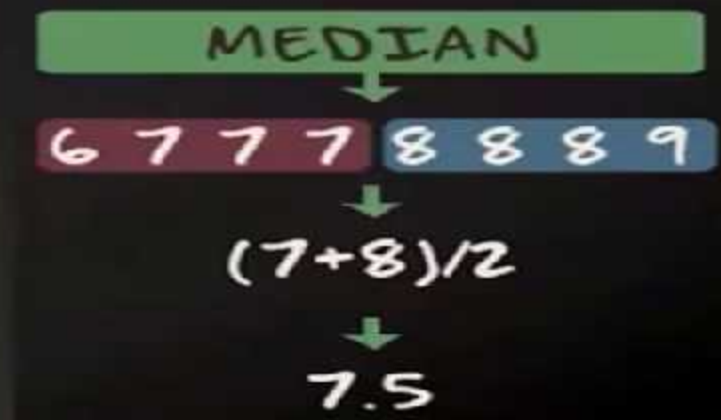




How sympathetic do you think Tomás Bayez is?

0 1 2 3 4 5 6 7 8 9 10

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8
Respondent 8	7



Mean

Mean

$$\bar{x} = \frac{\sum x}{n}$$

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8

	Sympathy Bayez
Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8

$$\bar{x} = \frac{\sum x}{n}$$

Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8
	53

$$\bar{x} = \frac{\sum x}{n}$$

Respondent 1	8
Respondent 2	7
Respondent 3	9
Respondent 4	8
Respondent 5	7
Respondent 6	6
Respondent 7	8
	53

$$\frac{53}{7} = 7.6$$

summarizing a distribution

CENTER

mode

median

mean

WHEN should you report
WHICH measure of central tendency?

summarizing a distribution

CENTER

mode

median

mean

WHEN should you report
WHICH measure of central tendency?

measurement level of your variable

summarizing a distribution

CENTER

mode

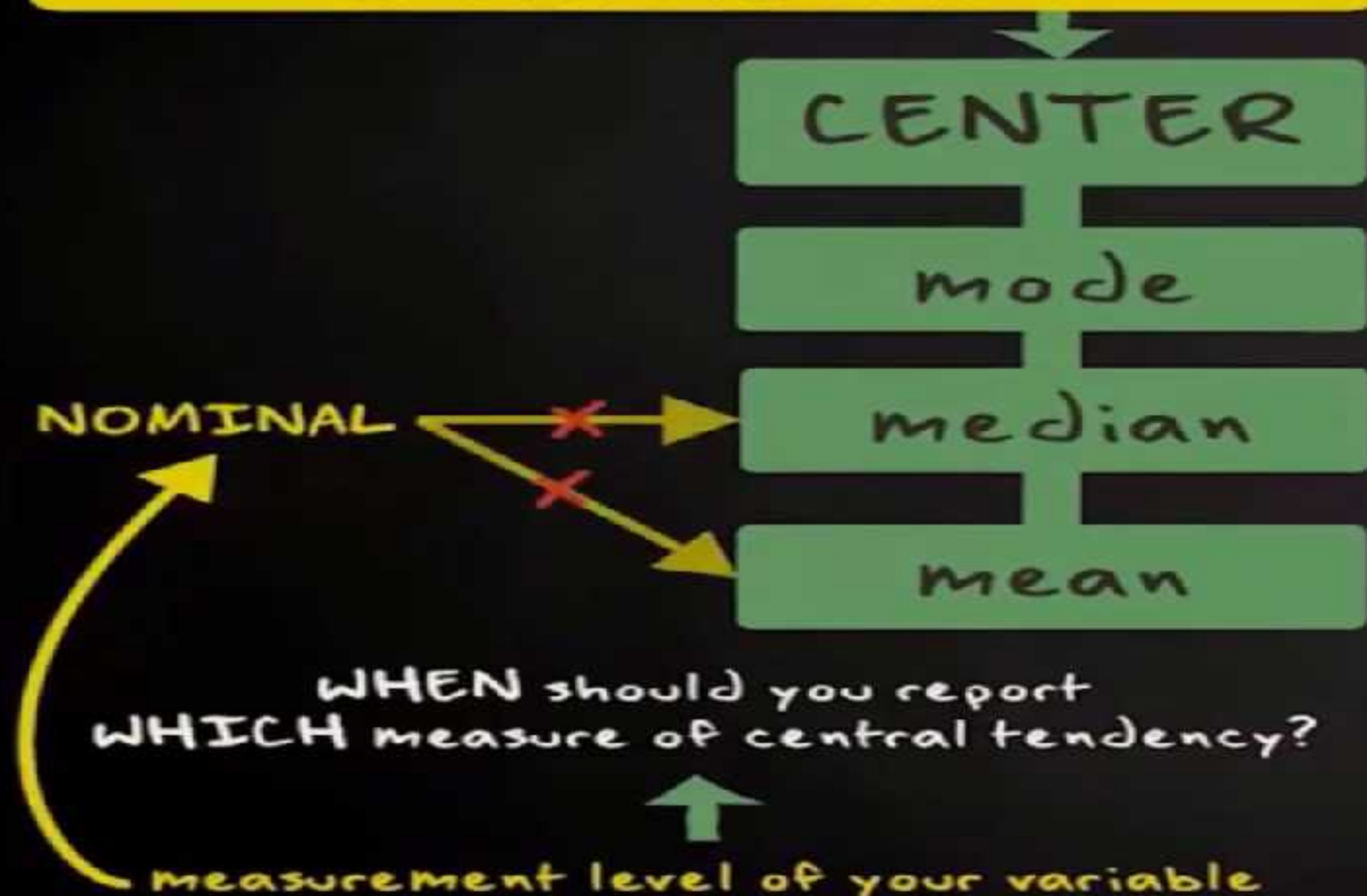
median

mean

NOMINAL

WHEN should you report
WHICH measure of central tendency?

measurement level of your variable



summarizing a distribution

CENTER

mode

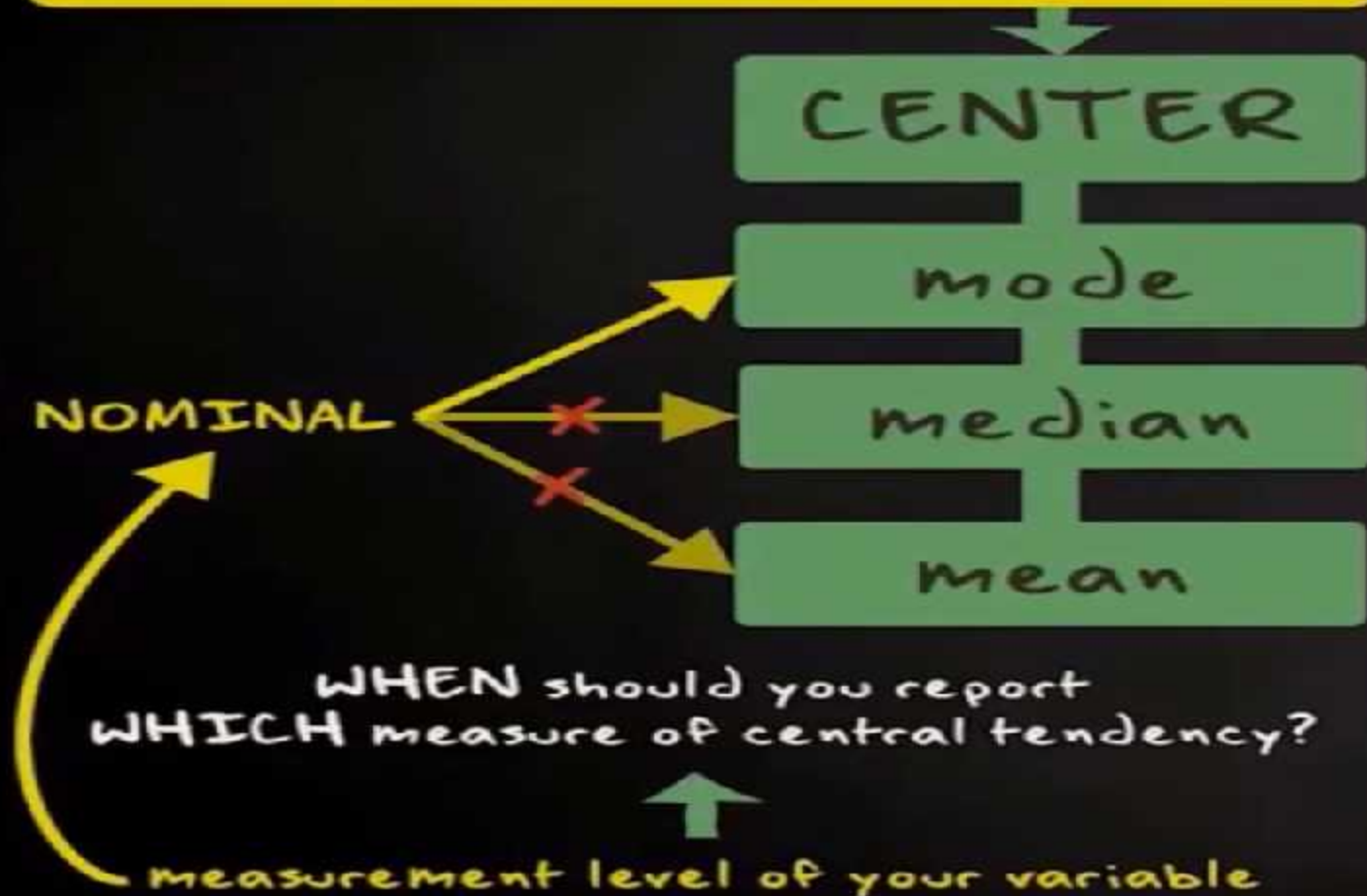
median

mean

NOMINAL

WHEN should you report
WHICH measure of central tendency?

measurement level of your variable



summarizing a distribution

CENTER

mode

median

mean

QUANTITATIVE?

WHEN should you report
WHICH measure of central tendency?

measurement level of your variable





	Income (in euros)
You	35.000
Bartender	25.000
Other guest 1	48.000
Other guest 2	41.000
Other guest 3	28.000
Other guest 4	37.000
Other guest 5	33.000



	Income (in euros)
You	35.000
Bartender	25.000
Other guest 1	48.000
Other guest 2	41.000
Other guest 3	28.000
Other guest 4	37.000
Other guest 5	33.000

$$\text{mean} = 35.286$$



	Income (in euros)
You	35.000
Bartender	25.000
Other guest 1	48.000
Other guest 2	41.000
Other guest 3	28.000
Other guest 4	37.000
Other guest 5	33.000

mean = 35.286

median = 35.000



	Income (in euros)
You	35.000
Bartender	25.000
Other guest 1	48.000
Other guest 2	41.000
Other guest 3	28.000
Other guest 4	37.000
Other guest 5	33.000

$\approx$ mean = 35.286
median = 35.000



	Income (in euros)
You	35.000
Bartender	25.000
Other guest 1	48.000
Other guest 2	41.000
Other guest 3	28.000
Other guest 4	37.000
Other guest 5	33.000

≈ mean = 35.286
median = 35.000



	Income (in euros)
You	35.000
Bartender	25.000
Other guest 1	48.000
Other guest 2	41.000
Other guest 3	28.000
Other guest 4	37.000
Other guest 5	33.000
Galtón	70.000.000



	Income (in euros)
You	35.000
Bartender	25.000
Other guest 1	48.000
Other guest 2	41.000
Other guest 3	28.000
Other guest 4	37.000
Other guest 5	33.000
Galtón	70.000.000

mean = 8.780.875

median = 36.000



	Income (in euros)
You	35.000
Bartender	25.000
Other guest 1	48.000
Other guest 2	41.000
Other guest 3	28.000
Other guest 4	37.000
Other guest 5	33.000
Galtón	70.000.000

~~mean = 37.000.075~~

median = 36.000



summary

summarizing
a distribution

CENTER

CATEGORICAL

mode

influential outliers/skewed
QUANTITATIVE

median

mean